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Injection site targeting device

Abstract

An apparatus for targeting an injection site using anatomical landmarks includes a first landmark identifier, a second landmark identifier, and a targeting band. The targeting band is connected with the first landmark identifier at a first end and a second landmark identifier at a second end. At least a portion of the targeting band is linearly deformable. In another embodiment, an apparatus for targeting an injection site using anatomical landmarks on a patient includes a targeting band having a first end and a second end, at least a portion of the targeting band being linearly deformable, a first landmark identifier pivotably connected to the targeting band at the first end, a second landmark identifier configured to be connected with the second end of the targeting band; and a fastener configured to secure the first and second landmark identifiers to the patient.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 62/281,496, filed Jan. 21, 2016, the entire disclosure of which is incorporated by reference herein.

FIELD OF THE INVENTION

(1) The present invention relates to an apparatus for targeting an injection site.

BACKGROUND

(2) In medical procedures, it can be important to administer an injection or place a treatment device at a specific site on the body. Nerve blocks, for example, can be used to manage, treat, and prevent pain. The efficacy of a nerve block, however, depends largely upon the ability of a health care professional to correctly identify the relevant nerve(s) and the injection site(s) associated with the relevant nerve(s), and to administer the appropriate pharmaceuticals in close proximity to the identified nerve without encountering structures, such as blood vessels, that are hidden beneath the skin.

(3) Nerves are typically located using anatomical landmarks, motor-evoked potential, or ultrasound. Each nerve-finding technique has advantages and drawbacks. However, they all generally require a significant amount of training beyond that of a general practitioner. Further, nerve finding via motor-evoked potential or ultrasound also requires costly equipment that requires additional training and may not be available in all clinical settings. Thus, there exists a need for a device that facilitates the administration of nerve blocks by eliminating the “guess work” associated with identifying injection sites. Further, the device increases the number of medical professionals, such as doctors of different specialties, and the like, that can administer nerve blocks in a clinical setting.

SUMMARY OF THE INVENTION

(4) The present invention provides an apparatus for targeting an injection site using anatomical landmark guidance. In accordance with one embodiment of the present invention, an apparatus for targeting an injection site using anatomical landmarks includes a first landmark identifier, a second landmark identifier, and a targeting band connected with the first landmark identifier at a first end and the second landmark identifier at a second end. At least a portion of said targeting band is linearly deformable.

(5) In accordance with another embodiment of the invention, an apparatus for targeting an injection site using anatomical landmarks on a patient includes a targeting band having a first end and a second end; a first landmark identifier pivotably connected to the targeting band at the first end; a second landmark identifier configured to be connected with the second end of the targeting band; and a fastener configured to secure the first and second landmark identifiers to the patient. At least a portion of the targeting is linearly deformable;

(6) In accordance with another embodiment of the invention, an apparatus for targeting an injection site using anatomical landmarks on a patient, includes a first landmark identifier configured to be fixed to the patient; a pivot arm having a pivotable connection with the first landmark identifier; a

targeting band having a first end and a second end, at least a portion of the targeting being linearly deformable, the first end of the targeting band being secured at the pivotable connection; and a second landmark identifier connected with the pivot arm.

(7) In accordance with yet another embodiment of the invention, 22. An apparatus for targeting an injection site using anatomical landmarks on a patient includes a targeting band having a first end and a second end; a landmark identifier having directional markers thereon; and a tensioning member having a first end and a second end, the tensioning member being pivotably connected to the landmark identifier at the first end, the tensioning member comprising an adjustable tensioning mechanism configured to move between the first end and the second end, and a marker to indicate a targeted location. The first end of the targeting band is connected with the first end of the tensioning member and the second end of the targeting band is connected with the adjustable tensioning mechanism.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1B illustrate an injection site targeting device in accordance with the present invention.
- (2) FIGS. 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I and 2J illustrate injection site targeting devices in accordance with the present invention.
- (3) FIG. 3 illustrates an injection site targeting device in accordance with the present invention.
- (4) FIG. 4 illustrates a base component of an injection site targeting device in accordance with the present invention.
- (5) FIG. 5 illustrates a pivot arm of an injection site targeting device in accordance with the present invention.
- (6) FIG. 6 illustrates a cross-section of an assembled injection site targeting device in accordance with the present invention.
- (7) FIG. 7 illustrates an assembled injection site targeting device in accordance with the present invention.
- (8) FIGS. 8A and 8B illustrate an injection site targeting device having a locking pivot arm, in accordance with the present invention.
- (9) FIGS. 9A and 9B illustrate an injection site targeting device having a pivot arm with a track and additional support, in accordance with the present invention.
- (10) FIGS. 10A and 10B illustrate an injection site targeting device having a pivot arm with a track, additional support, and configured to locate a landmark via manual palpation, in accordance with the present invention.

DETAILED DESCRIPTION

(11) Reference throughout the specification to “one embodiment,” “another embodiment,” “an embodiment,” “some embodiments,” and so forth, means that a particular element (e.g., feature, structure, property, and/or characteristic) described in connection with the embodiment is included in at least one embodiment described herein, and may or may not be present in other embodiments. In addition, it is to be understood that the described element(s) may be combined in any suitable manner in the various embodiments. Further, the term “user” herein describes an individual or group of individuals administering the injection. It should also be appreciated that the features disclosed herein may be combined in various ways without departing from the scope of the invention.

(12) The location of underlying structures and/or tissues in the body such as nerves, tendons, muscles, vasculature, spine structures such as discs and facets, bones, joints and organs, etc. can sometimes be determined relative to anatomical landmarks that are identifiable via palpation or

visual inspection. In some cases, the location of these underlying structures is well established via, for example, scientific study. For example, Levin identifies the location of the preferred injection site for a greater occipital nerve block as being located “approximately two thirds of the distance on a line drawn from the center of the mastoid to the external occipital protuberance.” Levin, Morris. “Nerve Blocks in the Treatment of Headache.” *Neurotherapeutics: The Journal of the American Society for Experimental NeuroTherapeutics* 7.1 (2010): 197-203, 198. Print. In some cases, literature suggests multiple approaches to locate a preferred injection site. Loukas, for example, suggests that the preferred injection site for a greater occipital nerve block is 2 cm lateral and 2 cm inferior to the greater occipital protuberance. Loukas, M., et al. “Identification of greater occipital nerve landmarks for the treatment of occipital neuralgia.” *Folia Morphologica* 65.4 (2006): 337-342, 347. Print. Still other publications suggest multiple injection sites for a greater occipital nerve block, including injection sites at one third the distance between the occipital protuberance and the mastoid process, at one half (the midpoint) the distance between the occipital protuberance and the mastoid process, and at a point 1-2 cm inferior to the midpoint. Voight, Crystal L. and Maurice O. Murphy. “Occipital Nerve Blocks in the Treatment of Headaches: Safety and Efficacy.” *The Journal of Emergency Medicine* 48.1 (2015): 115-129, 122. Print. Thus, it is essential that an injection targeting device be adaptable to a specific patient in order to permit a practitioner to administer an injection at a site using the landmark(s) and location methodology that the practitioner is familiar and comfortable with.

(13) In other cases, such as when a patient has known anatomical variations, the location of underlying structures and/or tissues may be determined for a specific patient via established methods including, but not limited to, imaging methods such as ultrasound, computed tomography (CT), and magnetic resonance imaging (MRI), nerve stimulation such as motor evoked potential testing, and palpation and percussion techniques. In these cases, preferred injection sites can be defined for a specific patient, for example by defining a preferred injection site relative to multiple (two or more) anatomical landmarks, or by defining an injection site in terms of a distance and direction (in terms of degrees, watch face, etc.) relative to one anatomical landmark. Once preferred injection sites are established, they can be noted in a patient's medical records so that testing does not need to be repeated.

(14) The present invention uses at least one anatomical landmark and known or established relationships relative to the at least one anatomical landmark to provide an injection site targeting device that facilitates consistent, reproducible administration of injections or other therapeutic means, such as nerve blocks, TENS, administration of medications, etc. across a variety of populations without requiring expensive diagnostic equipment. Additionally, because the injection site targeting device essentially serves as a template for the medical practitioner, the training necessary to administer a nerve block using the injection site targeting device can be reduced compared to the training required to administer nerve blocks using traditional anatomical landmark guidance techniques, imaging techniques, or nerve stimulation.

(15) Further, while the invention is described as an injection targeting device, it should be appreciated that the devices described herein are applicable to any treatment or procedure that relies on reliable identification of the location of nerves in order to be effective. Transcutaneous electrical nerve stimulation (TENS), for example, requires that electrodes be placed on the surface of the skin in close proximity to specific nerves. Any of the devices described herein could be adapted to identify a site for application or to mark the skin for repeated or later application, without departing from the scope of the invention. Further, while a nerve block is typically administered by a medical practitioner, it should be appreciated that, in some contexts, such as TENS administration, the devices described herein are appropriate for use by a lay person, and may be used to facilitate self-administration of treatment by a patient for the administration of therapy at home or in a non-clinical setting. The devices described herein could also be used to place a radiofrequency probe or catheter introducer at a specific location on the body for treatment,

diagnostics, or monitoring.

(16) FIGS. 1A and 1B illustrate an injection site targeting device **100** in accordance with the present invention. The injection site targeting device **100** includes a landmark identifier **110**, an adjustment mechanism **115** that includes a slider **120**, a targeting band **130**, a marker **140**, a measurement guide **150**, a directional guide **160**, and a connector **170**. The landmark identifier **110** is configured to be placed on or over an anatomical landmark that has been identified by a user. Landmark identifier **110** may be adapted in shape and size to suit, conform to, or accommodate the anatomical landmark it is intended to identify. For example, where the identified anatomical landmark is a bony landmark that protrudes from the body, such as the mastoid process, the occipital protuberance, the medial malleolus, or the lateral malleolus, the bottom of the landmark identifier may be concave or cupped to receive a bony projection, may be soft and moldable to conform to the bony landmark, or may include a partial depth hole or through hole that permits a user to identify the position of the landmark via visual inspection or palpation. A landmark identifier intended for use with the medial malleolus may be larger in diameter and have a deeper cup or concavity to its shape than a landmark identifier intended for the occipital protuberance. Where the chosen anatomical landmark does not protrude from the body, for example, in the case of soft tissue landmarks such as the edge of a facial feature such as an eyebrow, or underlying tissue that can be identified via palpation, the landmark identifier may have a flat bottom. In still other embodiments, landmarks may be identified via conventional imaging means and temporary, semi-permanent, or permanent marks can be made on a patient's skin to facilitate repetitive placement of a landmark identifier via visual inspection. In some cases, at least a portion of the landmark identifier may be inserted beneath the skin to facilitate identification of a specific landmark via physical inspection or palpation. Landmark identifier **110** may be constructed of any material, but are preferably constructed of a material or combination of materials that is unlikely to cause skin irritation. In some embodiments, it is preferable that the material or materials withstand chemical and/or heat sterilization procedures. Exemplary materials include, but are not limited to, polymers, including silicone polymers and plastics, and metals and metal alloys.

(17) Landmark identifier **110** may be held in place by a person, secured to the skin via a skin adhesive such as a silicone adhesive, secured via an elastic band, hook and loop fastener, tape, or any other fastener, frame, or combination thereof that allows for landmark identifier **110** to be held securely against the patient's body and the anatomical landmark. It should also be appreciated that, in some embodiments, the landmark identifiers may be labeled or color coded to indicate the anatomical landmark intended to be used with that specific landmark identifier. In some embodiments, landmark identifier **110** includes a directional guide **160**, indicating direction via, for example, degrees (360) or in terms of a clock face. In other embodiments, the landmark identifier may include an opening or port that allows a user to view and/or palpate an anatomical landmark to facilitate placement of the injection targeting device, or to view and/or palpate the anatomical landmark while the injection site targeting device is in place on a patient. Landmark identifier **110** may be constructed of any material suitable to be placed against a patient's skin, but is preferably a material unlikely to cause skin irritation on contact. In some embodiments, it is desirable for the material or combination of materials to be sterilizable. Exemplary materials include, but are not limited to, metals such as austenitic 316 stainless steel, martensitic 440 and 420 stainless steels, titanium alloys (such as Ti6Al4V), and the like, and polymers such as polypropylene, polytetrafluoroethylene, polyether ether ketone (PEEK), silicone, and the like.

(18) The adjustment mechanism **115** is connected to landmark identifier **110** via connector **170**. Connector **170** may be any fastener including, but not limited to, a pin, screw, nut, nut/bolt combination, clamp, adhesive, or any other fastener that can secure adjustment mechanism **115** to landmark identifier **110** permanently, temporarily, or removably. In some embodiments, the connector may be formed integrally with the landmark identifier and targeting band. In some embodiments, connector **170** permits the adjustment mechanism to pivot relative to the landmark

identifier. Adjustment mechanism **115** includes slider **120**, which is configured to move along the length of the adjustment mechanism **115**. In some embodiments, slider **120** can be secured in a specific position, for example, by tightening connector **175**. The length of adjustment mechanism **115** is such that it permits the slider to move at least the distance from the landmark locator to the desired injection site. Preferably the travel of the slider is greater than the distance from the landmark locator to the desired injection site. Further, the adjustment mechanism may be straight (flat), or curved to mimic the curvature of the part of the body on which it is to be used. The adjustment mechanism may be constructed of any material with sufficient structural characteristics to withstand manipulation and tension without deforming. In some embodiments, it is preferable that the adjustment mechanism be constructed of sterilizable metal or polymers, or a combination thereof, as discussed with respect to the landmark identifier.

(19) Targeting band **130** is connected to the adjustment mechanism at two points, and should be under increasing tension as slider **120** is moved away from landmark identifier **110**. At a first end, targeting band **130** is connected to the adjustment mechanism **115** and the landmark identifier **110** via connector **170** described above, or may be directly connected to adjustment mechanism **115**. In some embodiments, the connection between the landmark identifier and the targeting band is a pivotable connection. At a second end, targeting band **130** is connected to slider **120** via connector **175**. Connector **175** may be a pin, screw, bolt, nut/bolt combination, clamp, adhesive, or any other appropriate fastener or fixative to temporarily, permanently, or removably connect targeting band **130** with slider **120**. In some implementations, the targeting band may be secured at the second end by fixing the targeting band between 2 portions of the slider.

(20) Marker **140** is located on targeting band **130**, and is used to indicate the preferred injection site. The marker **140** may be located at any point on the targeting band that will extend beyond the boundaries of the landmark identifier under tension. In some embodiments, specific points are marked along the band, such as $\frac{1}{3}$, $\frac{2}{3}$, or both $\frac{1}{3}$ and $\frac{2}{3}$. In other embodiments, multiple markers may be located on the targeting band to define a scale. Targeting band **130** is deformable so that the position of marker **140** can be adjusted by moving slider **120** until marker **140** is in the desired position, and then securing the slider to the adjustment mechanism via, for example, a thumb screw. Adjustment mechanism **115** optionally includes measurement guide **150**, which may be a ruler or provide other consistent markings to indicate position relative to the identified anatomical landmark.

(21) In the embodiment illustrated in FIGS. 1A and 1B, targeting band **130** may be constructed of any material that will deform in the axial direction under the tension created as the slider **120** is moved away from landmark identifier **110**. Preferably, the material deforms linearly in the axial direction over a distance of at least a few millimeters to a few centimeters or more, depending on the distance from the landmark to the injection site, or the distance between the landmarks. In some embodiments, for example where the distance between landmarks is relatively large (50 or more centimeters), it is preferable that the material deforms linearly in the axial direction over at least a portion of the distance between the landmarks. For example, in dealing with nerves on the head the material should deform linearly in the axial direction when stretched 3-10 centimeters. This permits the same band to be used on patients of different shapes and sizes, and accounts for the variation in anatomical features that exist from person to person or patient to patient. When landmarks are more than a few centimeters apart, it may be preferable that only a portion of the targeting band immediately around the injection site be constructed of a material that deforms linearly in the axial direction, while the surrounding sections of the targeting band are constructed of a material that does undergoes minimal or no deformation in the axial direction. In some embodiments, for example, where only one adjustment is necessary, such as embodiments where the targeting band is intended to be disposable or for single use, plastic deformation of targeting band **130** is acceptable. A non-limiting example of a material that would undergo plastic deformation in this context is polyvinyl chloride (plastic wrap). In other embodiments, for example, where repeated adjustment

may be necessary, or where the targeting band is to be reused, elastic deformation is preferable. A non-limiting example of a material that deforms linearly in this context is high purity silicone rubber.

(22) Marker **140** indicates the desired injection site and may be a mark on the targeting band, a through hole, a through hole covered by a thin film, a scored mark or partial-depth hole configured to guide a needle and/or limit the depth to which a needle can be inserted, or a tapered or funnel-shaped opening configured to guide a needle on the superior or inferior surface, direct the trajectory of a needle, and/or limit the depth to which a needle can be inserted. In some embodiments, marker **140** may be formed continuously as part of targeting band **130**, for example, when targeting band **130** is formed via a molding process using high purity silicone rubber. In other embodiments, marker **140** may be formed of a different material, such as metal or plastic, that is inserted into an opening in targeting band **130** or fixed in targeting band **130** during the manufacturing process, forming an eyelet or tunnel in the targeting band. In some embodiments, a marker may be added by a user after manufacture, for example by piercing the targeting band at an appropriate point and inserting a metal or plastic marker, or by making a mark on the targeting band before use, for example, with a permanent marker.

(23) The embodiment illustrated in FIGS. **1A** and **1B** is particularly useful in cases such as that described by Loukas, where the preferred injection site for a greater occipital nerve block is indicated as being 2 cm medial and 2 cm inferior to the occipital protuberance. Using basic geometry, the preferred injection site is approximately 2.83 cm from the occipital protuberance at about 270 degrees. In order to use the injection targeting device of FIGS. **1A** and **B**, one would place the landmark identifier on a patient's occipital protuberance, orienting the direction guide in the same manner as a compass, adjust and fix the slider so that the marker is located 2.83 cm from the landmark identifier, and pivot the adjustment mechanism (and targeting band) so that the marker is located at 270 degrees using the direction guide.

(24) FIGS. **2A** and **2B** illustrates an injection site targeting device in accordance with the present invention. The injection site targeting device includes two landmark identifiers **210**, **220**, a targeting band **230**, connectors **270** and **275**, and a marker **240** to identify a preferred injection site. Each landmark identifier **210**, **220** is configured to be placed on an anatomical landmark that has been identified by a medical professional or by a layperson. Landmark identifiers **210** and **220** may be adapted in shape and size to suit the landmark they are intended to identify without departing from the scope of the invention. For example, where the identified anatomical landmark is a bony landmark that protrudes from the body, such as the mastoid process, the occipital protuberance, the medial malleolus, or the lateral malleolus, the bottom of the landmark identifier may be concave or cupped to receive a bony projection, may be soft and moldable to conform to the bony landmark, or may include a partial depth hole or through hole to permit visual inspection and/or palpation by a user. A landmark identifier intended for use with the medial malleolus may be larger than a landmark identifier intended for the occipital protuberance. Where the chosen anatomical landmark does not protrude from the body, for example, in the case of soft tissue landmarks such as the edge of a facial feature such as an eyebrow, the landmark identifier may have a flat bottom.

(25) Landmark identifiers **210** and **220** may be constructed of any material, but are preferably constructed of a material or combination of materials that is unlikely to cause skin irritation. In some embodiments, it is preferable that the material(s) can be sterilized. Exemplary types of materials include, but are not limited to, polymers, including plastics and silicone, and metals and metal alloys. Exemplary materials include, but are not limited to, metals such as austenitic 316 stainless steel, martensitic 440 and 420 stainless steels, titanium alloys (such as Ti6Al4V), and the like, silicone, and plastics such as polypropylene, polytetrafluoroethylene, polyether ether ketone (PEEK), and the like. Landmark identifiers **210** and **220** may be held in place by a person, secured via a skin adhesive such as a silicone adhesive, secured via an elastic band, hook and loop fastener, tape, deformable materials, or any other fastener, frame, or combination thereof that allows for

landmark identifiers **210** and **220** to be held securely against the patient's skin and, in turn, the identified landmark. It should also be appreciated that, in some embodiments, the landmark identifiers may be labeled or color coded to indicate the anatomical landmark intended to be used with that specific landmark identifier.

(26) Targeting band **230** is connected to the landmark identifiers **210**, **220** via connectors **270**, **275**. Connectors include, but are not limited to, pins, adhesives, clamps, screws, nuts, bolts, and the like. In some embodiments, the landmark identifiers may be formed integrally with the targeting band. The connection between targeting band **230** and the landmark identifiers may be permanent or removable and may be pivotable/rotatable or fixed. Targeting band **230** may be constructed of any material that deforms linearly under slight tension. In some embodiments, the targeting band may be constructed of a material that undergoes linear elastic deformation under slight tension. Preferably, targeting band **230** is constructed of a material that undergoes linear deformation under slight tension. In some embodiments, targeting band **230** may undergo linear elastic deformation under slight tension.

(27) In some embodiments, the targeting band is constructed of a material that has a type A durometer hardness of 50-80, a tensile strength of 1000-1500 psi, elongation of 300%-700%, tear strength of 100-400 ppi, and specific gravity of 1-1.4. Exemplary materials include, but are not limited to, medical grade silicone including high purity silicone rubber, polyurethane, urethane copolymers, and foamed silicone. In some embodiments, it is preferable that the targeting band be constructed of an auxetic material, which has a negative Poisson's ratio in addition to the aforementioned materials characteristics, meaning that the material becomes slightly thicker and, in some cases, more rigid when stretched. In some embodiments, it is preferable, that the targeting band be constructed of an elastomer (such as polyurethane or silicone materials) that has a durometer hardness of 50-70 Shore A, a tensile strength of 1100-1300 psi, elongation of 300-500%, tear strength of 200-300 ppi. The targeting band is preferably slightly shorter than the distance between the two landmarks (in average populations) so that, when properly positioned, the targeting band is under slight tension. In still other embodiments, the targeting band may be trimmed to a desired length by a user.

(28) Marker **240** marks the preferred injection site on the targeting band, based on the relative relationship between the landmarks. The location of marker **240** is determined based on the location of the desired injection site (nerve) relative to two anatomical landmarks, for example 40% of the distance between landmark A and landmark B. Thus, the location of the marker **240** may be varied based on the desired injection site and its position relative to anatomical landmarks. In some embodiments, the marker is a dot or other symbol printed on the targeting band. In other embodiments, the marker is a through hole or partial-depth hole that is covered by a thin membrane or film that is configured to be punctured by a needle, for example to indicate that the targeting band is "new." In still other embodiments, marker **240** is a shaped opening that serves as a needle guide and may be configured to limit the depth of insertion of the needle by forming a physical barrier, and/or may be configured to guide the needle at a specific path or trajectory. As described with respect to the embodiment illustrated in FIGS. **1A** and **1B**, the marker may be formed of the same material as the targeting band via, for example, molding, or may be formed of another material entirely and inserted in the targeting band during or after manufacture. In still other embodiments, the marker may be a scale printed on the targeting band or series of holes in the targeting band at defined intervals. By providing relative measurements rather than a specific mark, a medical professional can use his or her knowledge to choose an appropriate injection site using the scale. Because the targeting band undergoes linear deformation under slight tension, marker **240** will move as the targeting band stretches. Thus, a marker located 40% between the two landmark identifiers will continue to remain at 40% as the landmark locators are moved apart and the targeting band is stretched. This linear deformation allows the injection targeting device to be used on patients of varying shapes and sizes with consistent and reproducible results.

(29) FIGS. 2C and 2D illustrate exemplary embodiments of injection targeting devices where the landmark identifiers **210** and **220** are formed integrally with the targeting band **230**, and designed to be held in place using an adhesive and/or pressure applied by a practitioner, provider, the patient, or a third party. In FIG. 2C, landmark identifiers **210** and **220** include through holes configured to allow a user to inspect the patient's body, for example via visual inspection or manual palpation, in order to facilitate identification of the desired landmarks. FIG. 2D illustrates an embodiment similar to that illustrated in FIG. 2C, but includes integrated personal protective equipment (PPE) **225**, in this case a glove, is connected with the landmark identifiers **210**, **220** at 2 points. While the illustrated connection points to the landmark identifier is illustrated at the thumb and forefinger of the glove, it should be appreciated that the PPE may be connected at any points that will allow for the medical practitioner to properly locate the landmark identifiers while placing tension on the targeting band. The connection points may vary, for example, with the size of the glove. A glove designed for a larger hand may connect at the thumb and index finger, while a glove designed for a smaller hand may connect at the thumb and middle finger.

(30) FIGS. 2E and 2F illustrate two different views of an exemplary embodiment of an injection targeting device configured to conform to the body of a patient. In the embodiment of FIGS. 2E and 2F, targeting band **230** has an inverted v-shape. Targeting band **230** includes multiple markers **240**. Landmark identifiers **210** and **220** include through holes, and are formed integrally with targeting band **230**. Targeting band **230** is attached to or embedded in a fastener **270**, which is an elastic material that stretches and conforms to a patient's body to secure the landmark identifiers and, thus, markers in place on a patient. In some embodiments, the elastic material may be secured in place via hook and loop fasteners, snaps, buckles, or the like. In other embodiments, the elastic material may be self adhering material, such as 3M™ Coban™, that is secured by overlapping the material once it is wrapped around the body. Preferably, if the material is to be wrapped around the face or a joint, openings exist in the material to accommodate facial features, ears, and other anatomical features. The embodiment illustrated in FIGS. 2E and 2F also permits for simultaneous identification of multiple injection sites in the case where bilateral injection sites exist on the body.

(31) FIGS. 2G and 2H illustrate an exemplary implementation in which a frame is used to fix the targeting band to a patient. Specifically, the fastener **270** in this case is an adjustable frame that fits behind the head and over and around the ears of a patient, similar to “behind the head headphone” designs. Once the occipital protuberance **220** is located, the rear band of the frame can be adjusted to fit the head of the patient, for example via spring tension, sliding, or ratcheting mechanisms, or the like. The ends of the rear band are semi-rigid and are configured to be bent around the ears of the patient, in the reverse fashion of conventional eyeglasses. For example, the rear band of the frame extends over the top of a patient's ears, and down along the front of the ear, and each end of the band is bendable around the bottom of the patient's ears and back towards the mastoid process. The semi-rigid, bendable/deformable ends of the band also serve as landmark identifiers **210**. Targeting band **230** is connected to the fastener **270** via a pivoting connection **255**, which allows the targeting band to be secured to landmark identifier **210** on either side of the patient's head via a fastener, adhesive, or the like. Once targeting band **230** is secured to the second landmark identifier **210**—identifying the mastoid process on either side of the patient—the targeting band **230** is under tension and the markers **240** can be used to identify injection sites. The rear band can be formed of a polymer, plastic, or metal. The semi-rigid (deformable) portions should retain the new shape once positioned, accordingly it is preferable that the semi-rigid portions be constructed of a deformable metal interior (wire) covered by foam, fabric, or other soft material that is unlikely to cause skin irritation, for patient comfort.

(32) FIGS. 2I and 2J illustrate an exemplary implementation in which a combination of a frame and band used to fix the targeting band to a patient. In this embodiment, the fastener **250** is an adjustable frame that fits behind the head and sits on top of a patient's ears. A band **270**, such as an elastic band or hook and loop band, is configured to wrap around the front of a patient's head to

further secure the frame. Once the occipital protuberance **220** is located, the rear band of the frame can be adjusted to fit the head of the patient, for example via spring tension, sliding, or ratcheting mechanisms, or the like. The ends of the rear band are semi-rigid and are configured to be bent along the back of a patient's ears, similar to the earpieces of conventional eyeglasses. The semirigid, bendable/deformable ends of the band also serve as landmark identifiers **210**. Targeting band **230** is connected to the fastener **250** via a pivoting connection **255**, which allows the targeting band to be secured to landmark identifier **210** on either side of the patient's head via a fastener, adhesive, or the like. Once targeting band **230** is secured to the second landmark identifier **210**—identifying the mastoid process on either side of the patient—the targeting band **230** is under tension and the markers **240** can be used to identify injection sites. The rear band can be formed of a polymer, plastic, or metal. The semi-rigid (deformable) portions should retain the new shape once positioned, accordingly it is preferable that the semi-rigid portions be constructed of a deformable metal interior (wire) covered by foam or other soft material that is unlikely to cause skin irritation, for patient comfort.

(33) To use the injection targeting device of FIGS. 2A through 2J, a medical professional locates the first anatomical landmark and fixes the first landmark identifier in place. The medical professional then fixes the second landmark identifier in place, using a fastening mechanism, and placing slight tension on the targeting band. The marker should then be in position to identify the preferred injection site so that a nerve block can be administered because the targeting band stretches linearly.

(34) FIG. 3 illustrates an injection site targeting device in accordance with the present invention. In the embodiment illustrated in FIG. 3, the first landmark identifier **310** is configured to be anchored to a patient by elastic band **360**. It should be appreciated, however, that any of the fixation mechanisms described herein may be used to anchor the injection site targeting device including, but not limited to, hook and loop fasteners, adhesive, a string, a headband, etc. The first landmark identifier **310** and the second landmark identifier **320** are connected via a ratcheting tensioning member **350**, and targeting band **330** is disposed within ratcheting tensioning member **350** such that the ratcheting tensioning member tensions the targeting band **330** as the position of the second landmark identifier **320** is adjusted. The tensioning mechanism may be constructed of materials such as those described with respect to the adjustment mechanism of FIGS. 1A and 1B, and the targeting band is preferably constructed of a material as described with respect to the embodiment illustrated in FIGS. 2A and 2B. While FIG. 3 illustrates a ratchet mechanism to adjust the length of the tensioning member, it should be appreciated that a variety of mechanical mechanisms could be used including, but not limited to, a cam mechanism, a slide, a threaded device, or the like could be used to adjust the tensioning member.

(35) FIG. 4 illustrates a base component of an injection targeting device designed for the abortive and prophylactic treatment of headaches using occipital nerve blocks, which provides a pivotable connection to a pivot arm of an injection targeting device in accordance with the present invention. However, it should be appreciated that the design can be adapted to work with any injection site that can be defined by its position relative to an anatomical landmark. In the embodiment illustrated in FIG. 4, attachment points **460** include loops through which an elastic band, a ribbon or string, a cam or ratcheting strap, or a length of hook and loop fastener can be threaded to affix the device to a patient. It should be appreciated that the base component can be held in place using any of the means described herein. Landmark locator **410** is located at the bottom of the base component, and includes a concavity or depression to accommodate a patient's occipital protuberance. Threaded rod **470** protrudes from the base component through a slot **480** in rotary mount **420**, and is pivotably connected to the base component, so that threaded rod **470** pivots in slot **480** in rotary mount **420**. Pivot or rotary mount **420** allows for rotation of the threaded rod **470**, so that threaded rod can pivot in the desired direction or orientation to accommodate curvature of a body part such as the skull. In some embodiments, it is preferable that the base component is comprised of metal or

another material capable of withstanding sterilization, as described with respect to the embodiments in FIGS. 1A, 1B, 2A, and 2B.

(36) FIG. 5 illustrates a pivot arm including a tensioning member and a targeting band in accordance with the present invention, for use with the base component illustrated in FIG. 4. Tensioning member **415** is a mechanical slide that is curved to accommodate the shape of a human head. As illustrated in FIG. 5, targeting band **430** is sandwiched between the tensioning member **415** and landmark locator **420** at first end. However, various configurations for the tensioning member are possible. For example, the tensioning member may extend from the side of the device, to facilitate manipulation of the tensioning member and minimize interference with the targeting band and injection site location. It should also be appreciated that the targeting band may be attached to the tensioning member using any conventional attachment mechanism. Targeting band **430** has a second attachment point **435** at its second end, which is configured to receive threaded pivoting rod **470**. When the injection targeting device is completely assembled, threaded pivoting rod **470** extends through second attachment point **435** in targeting band **430** and through sliding retaining bracket **460**. The base component/targeting band/sliding retaining bracket assembly is secured together via thumb nut **490**. Sliding retaining bracket **460** slides within the slot of tensioning member **415** when tension applied via thumb nut **490** permits for adjustment. When sliding retaining bracket **460** is in the desired position, which preferably places landmark locators **410** and **420** over the desired anatomical landmarks, thumb nut **490** can be tightened to fix the injection targeting device in the appropriate position to administer an injection. It should be appreciated that the materials for the base member, tensioning member, landmark identifiers, and the targeting band have been described with respect to the embodiments illustrated in FIGS. 2A, 2B, and 3.

(37) Landmark locator **420** has a concave bottom configured to receive the mastoid process. Once landmark locator **420** is placed, thumb nut **490** is tightened to fix the retaining bracket **460** in position, placing tension on targeting band **430**, which deforms linearly, as described with respect to the embodiment illustrated in FIGS. 2A and 2B, so that the marker remains in the same position relative to the landmark locators. The fully assembled injection targeting device is illustrated in FIGS. 6 and 7. Markers **440** and **450** identify the target injection sites for the lesser occipital nerve and greater occipital nerve, respectively. One known set of guidelines for locating an injection site for greater occipital nerve blocks and lesser occipital nerve blocks provides that the lesser occipital nerve is located one third the distance starting at the mastoid process and traveling in a line to the occipital protuberance, and the greater occipital nerve is located one third the distance starting at the occipital protuberance and traveling in line to the mastoid process. Thus, in the case of the greater occipital nerve and lesser occipital nerve, two injection sites can be targeted simultaneously with one device.

(38) It should be appreciated that the marker may be placed at any specified or desired point on the targeting band, and that multiple markers may be placed on the same band. In a non-limiting example, one occipital nerve block administration guideline suggests that a preferred injection is located at a distance of 20% to 23% from the occipital protuberance when traveling in line to the mastoid process. Thus, if this is a user's preferred guideline, a marker may be placed anywhere from 20% to 23% of the length of the targeting band, for example, a marker may be placed at 22% for this use.

(39) FIGS. 8A and 8B illustrate an injection site targeting device having a locking pivot arm, in accordance with the present invention including a linearly deformable targeting band. In the embodiment of FIGS. 8A and 8B, landmark identifier **820** is fastened to a landmark (in this case, the occipital protuberance) on a patient via a headband **850** constructed of elastic material, hook and loop fastener, or headband design that permits for adjustability in sizing. The pivot arm **880** is attached to landmark identifier **820** via a pivotable connection **855**. Pivotable connections include all pivot joints, including friction lock pivot joints, tension pivot joints, sliding pivot joints,

combinations thereof, and the like. In the embodiment illustrated in FIGS. **8A** and **8B**, the connection is also slidable, permitting manipulation of the pivot arm **880** by a user so that landmark identifier **810** can be placed on desired landmark (in this case the mastoid process). Any of a variety of mechanical locking mechanisms, such as a push button locking/unlocking mechanism, adjustable tension, or combinations thereof, may be used to permit manipulation of the pivot arm and/or lock the pivot arm in place. Targeting band **830**, including markers **840**, is fixed to the pivot arm at landmark identifier **810**, and at pivotable connection **855**. Thus, targeting band **830** deforms as pivot arm **880** is manipulated.

(40) FIGS. **9A** and **9B** illustrate an injection site targeting device having a pivot arm with a track and additional support, in accordance with the present invention including a linearly deformable targeting band. In the embodiment of FIGS. **9A** and **9B**, landmark identifier **920** is fastened to a landmark (in this case, the occipital protuberance) on a patient via a headband **950** constructed of elastic material, hook and loop fastener, or headband design that permits for adjustability in sizing. The pivot arm **980** is attached to landmark identifier **920** via a pivotable connection **955**. Pivotable connections include all pivot joints, including friction lock pivot joints, tension pivot joints, sliding pivot joints, combinations thereof, and the like. In the embodiment illustrated in FIGS. **9A** and **9B**, the connection is also slidable, permitting manipulation of the pivot arm **980** by a user so that landmark identifier **910** can be placed on desired landmark (in this case the mastoid process). Any of a variety of mechanical locking mechanisms, such as a push button locking/unlocking mechanism, adjustable tension, or combinations thereof, may be used to permit manipulation of the pivot arm and/or lock the pivot arm in place. The embodiment illustrated in FIGS. **9A** and **9B** also includes a support **985**, which is designed to provide additional support and stability for pivot arm **980** during the injection process.

(41) In the embodiment of FIGS. **9A** and **9B**, targeting band **930**, including markers **940**, is fixed to the pivot arm at landmark identifier **810**, and at pivotable connection **855**. Markers **940** include needle guides or eyelets, which provide structure for the markers. Pivot arm **980** includes a track **990**, which slidably guide the needle guides as targeting band **930** deforms linearly.

(42) FIGS. **10A** and **10B** illustrate an injection site targeting device having linearly deformable targeting band and a pivot arm with a track, additional support, and configured to locate a landmark via manual palpation, in accordance with the present invention. In the embodiment of FIGS. **10A** and **10B**, landmark identifier **1020** is fastened to a landmark (in this case, the occipital protuberance) on a patient via a headband **1050** constructed of elastic material, hook and loop fastener, or headband design that permits for adjustability in sizing. The pivot arm **1080** is attached to landmark identifier **1020** via a pivotable connection **1055**. Pivotable connections include all pivot joints, including friction lock pivot joints, tension pivot joints, sliding pivot joints, combinations thereof, and the like. In the embodiment of FIGS. **10A** and **10B**, landmark identifier **1010** is a through hole in targeting band **1030**. In some embodiments, the through hole may be reinforced with a rigid material such as a metal or plastic insert. Any of a variety of mechanical locking mechanisms, such as a push button locking/unlocking mechanism, adjustable tension, or combinations thereof, may be used to permit manipulation of the pivot arm and/or lock the pivot arm in place. Targeting band **1030**, including markers **1040**, is configured to slide or be stretched along a slot or track **1090** in pivot arm **1080**. In some embodiments, the position of targeting band **1030** and markers **1040** is secured by the user placing his or her finger on the landmark through the hole **1010**. In other embodiments, targeting band **1030** slides along a track that includes a fixation mechanism that squeezes or clamps the band in place once the second landmark is located, fixing the targeting band in place and maintain tension on the targeting band. In still other embodiments, the position of the targeting band in the track may be manipulated via a ratchet mechanism. Support **1085** at the free end of the pivot arm provides support for the injection targeting device when in use.

(43) It should be understood that, while examples are provided for targeting the greater and lesser

occipital nerves, the targeting device disclosed herein can be applied to consistently target any location on the body that has a location that can be described relative to one landmark with a distance and relative direction, or that can be described relative to two landmarks, without departing from the scope of the invention. All figures included herein should be interpreted as illustrative of non-limiting, exemplary embodiments, and not in a limiting sense.

Claims

1. An apparatus for targeting an injection site using anatomical landmarks, comprising: a curved frame configured to fit around the back of a human head; a first deformable arm extending from the curved frame; a first landmark identifier connected with the curved frame; a second landmark identifier being located along the first deformable arm and being configured to identify a mastoid process; and a first targeting band connected with the first landmark identifier at a first end and the second landmark identifier at a second end and having an injection site identified along said first targeting band, said targeting band being deformable and said injection site being positioned to identify a preferred location for a greater occipital nerve block when the second landmark identifier is positioned over the mastoid process.
2. An apparatus according to claim 1, wherein the first landmark identifier is connected with the curved frame at the center of the curved frame.
3. An apparatus according to claim 1 wherein the first landmark identifier is positioned at the center of the curved frame and formed integrally with the curved frame.
4. An apparatus according to claim 1, further comprising: a marker identifying the injection site on the first targeting band.
5. An apparatus according to claim 1, further comprising a second injection site identified along said first targeting band.
6. An apparatus according to claim 5, wherein said second injection site on said first targeting band is positioned to identify a location for a lesser occipital nerve block when the second landmark identifier is positioned over the mastoid process.
7. An apparatus according to claim 4, where the marker is a through hole.
8. An apparatus according to claim 5, further comprising: a marker identifying the second injection site.
9. An apparatus according to claim 8, wherein the marker is a through hole.
10. An apparatus according to claim 1, wherein the curved frame is an around the head-style headband.
11. An apparatus according to claim 1, further comprising: a second deformable arm extending from the curved frame; a third landmark identifier being located along the second deformable arm and being configured to identify a mastoid process; and a second targeting band connected with the first landmark identifier at a first end and the third landmark identifier at a second end and having an injection site identified along said second targeting band, said second targeting band being deformable and said injection site being positioned to identify a preferred location for a greater occipital nerve block when the third landmark identifier is positioned over the mastoid process.
12. An apparatus according to claim 11, further comprising: a marker identifying the first injection site on the second targeting band.
13. An apparatus according to claim 11, further comprising a second injection site identified along said second targeting band.
14. An apparatus according to claim 11, wherein said second injection site on said second targeting band is positioned to identify a location for a lesser occipital nerve block when the third landmark identifier is positioned over the mastoid process.
15. An apparatus according to claim 12, where the marker is a through hole.
16. An apparatus according to claim 15, further comprising: a marker identifying the second

injection site on the second targeting band.

17. An apparatus according to claim 15, wherein the marker is a through hole.
